

Observability-Aware Modem Selection in USV–AUV Navigation

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Abstract—This work investigates adaptive acoustic ranging for positioning in collaborative USV–AUV missions. We present an adaptive acoustic beacon management framework that dynamically selects a subset of available beacons to reduce acoustic energy while maintaining estimation quality. Our selector evaluates candidate beacon subsets using observability (Fisher Information), an explicit battery state-of-charge (SOC) model, and a mission uncertainty objective derived from a posterior covariance approximation. Validated in simulation with Monte Carlo runs, we achieved 49.9% reduced energy consumption versus a four beacon baseline, while preserving bounded performance (RMSE 0.85 m vs. 0.77 m).

Index Terms—AUV localization, acoustic positioning, EKF, sensor scheduling, Fisher Information Matrix, GDOP, energy-aware switching.

I. INTRODUCTION

Subsea navigation relies on multi-sensor fusion since absolute positioning is unavailable. This motivates multi-sensor fusion approaches (IMU, DVL, pressure) often combined with acoustic positioning (LBL/SBL/USBL) to estimate motion using Bayesian estimators including the Extended Kalman Filter (EKF). However, acoustic updates incur costs like latency, environmental sensitivity, beacon positioning, and energy consumption during long-duration missions [1].

Prior work has advanced underwater navigation through adaptive fusion and uncertainty modeling [2], [3], and has shown how geometry constrains localization accuracy via Fisher information and related bounds [4]. Separately, energy-aware designs acknowledge that acoustic updates cannot be used continuously [5]. Nevertheless, most approaches assume that acoustic sensing is either continuously available or configured statically, leaving limited guidance on how to manage modem activation as mission conditions evolve.

This work addresses this gap by introducing an observability-aware modem management approach for collaborative marine navigation. We implement an asynchronous EKF that fuses IMU, DVL, depth, and range-only acoustic measurements while explicitly maintaining uncertainty estimates. We then evaluate how activating different beacon subsets changes information content using observability checks and Fisher information-based metrics (GDOP/CRLB). The resulting framework supports mission-aware modem selection that maintains uncertainty requirements while reducing unnecessary acoustic activity.

II. METHODOLOGY

A. State Estimation and Measurements

We estimate AUV state x_t using an asynchronous Extended Kalman Filter (EKF) that predicts from inertial inputs and updates from velocity, depth, and acoustic range measurements. Acoustic ranges to beacon i at position b_i are:

$$r_i = \|p - b_i\| + v_i, \quad v_i \sim \mathcal{N}(0, \sigma_r^2), \quad (1)$$

with Jacobian H_i evaluated at predicted position. The standard EKF recursion combines predict and update steps with Kalman gain K_t .

B. Observability Metrics

For each candidate beacon subset S , we compute Fisher Information Matrix (FIM):

$$J_S = \sum_{i \in S} \frac{1}{\sigma_r^2} H_i^\top H_i, \quad (2)$$

and derive observability metrics: $\log \det(J_S)$ and GDOP.

C. Energy Model

Battery state-of-charge (SOC) evolves as:

$$P(n) = P_{\text{base}} + n P_{\text{beacon}}, \quad \text{SOC} = 1 - \frac{\text{Energy consumed}}{E_{\text{battery}}}, \quad (3)$$

where n is the number of active beacons. This enables energy-aware selection.

D. Adaptive Beacon Selection

At each decision epoch, we score feasible subsets as a weighted sum:

$$\text{Score}(S) = w_{\text{obs}} \cdot f_{\text{obs}}(S) + w_{\text{energy}} \cdot f_{\text{energy}}(S) + w_{\text{mission}} \cdot f_{\text{mission}}(S), \quad (4)$$

where:

- $f_{\text{obs}}(S)$ is normalized observability ($\log \det(J_S)$, ranging 0–1),
- $f_{\text{energy}}(S)$ combines remaining SOC and power draw normalized to [0,1],
- $f_{\text{mission}}(S)$ measures proximity to target uncertainty σ_{target} via posterior approximation,
- $w_{\text{obs}}, w_{\text{energy}}, w_{\text{mission}}$ are mission-phase weights.

The empty set $S = \emptyset$ (acoustics off) is allowed only if estimated uncertainty falls below the mission accuracy threshold, ensuring mission-defined accuracy bounds are not violated.

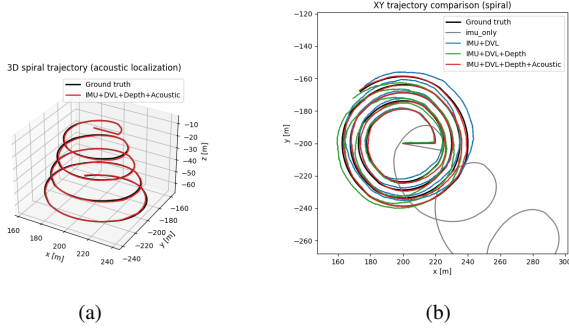


Fig. 1. Spiral trajectory in 3D and projected XY view, showing ground truth and EKF estimates for different sensor configurations.

To prevent switching chatter, we enforce hysteresis: switches occur only after minimum dwell time and when the new subset improves score by a margin, or saves energy without suffering from score degradation.

III. EXPERIMENTAL SETUP

Experiments are conducted in the HoloOcean simulator [6] using a collaborative USV–AUV scenario. The AUV executes a spiral trajectory to induce time-varying beacon geometry and test adaptive responsiveness. We conduct $N = 30$ Monte Carlo trials across three configurations: (1) baseline without acoustics, (2) continuous four beacon acoustics, and (3) proposed switching policy. For each trial, we randomize initial state error and sensor noise realizations while holding the trajectory fixed. Measured outputs include position RMSE, final position error, consistency metrics, and energy consumption. Each beacon is assigned a power rating of 10 Watts and P_{base} is set to zero for a pure acoustic comparison.

IV. RESULTS

Table I summarizes the Monte Carlo performance during the 300 second runs. The baseline exhibits a significant drift (RMSE 1.72 m), confirming that acoustics are necessary for bounded error. The four-beacon baseline achieves the best accuracy (RMSE 0.77 m; final error 0.47 m) but consumes the maximum acoustic energy (6.67 Wh). The proposed adaptive policy reduces acoustic energy by 49.9% (3.33 Wh) while maintaining moderate accuracy (RMSE 0.85 m). Importantly, the adaptive policy uses an average of 2 active beacons over time, indicating that the selector frequently turns acoustics off or selects small subsets, yet still retains recoverability when uncertainty rises. Consistency diagnostics further show that acoustic-update (Normalized Innovation Squared)NIS remains slightly higher for adaptive system (95.73%), than the four-beacon baseline (95.16%) suggesting the switching mechanism does not introduce unstable measurement gating. (Normalized Estimation Error Squared)NEES percentages on the other hand indicate partial inconsistency that motivates current tuning of noise models and gating thresholds.

TABLE I
MONTE CARLO RUNS SUMMARY. ENERGY CONSERVED REPORTED AS PERCENTAGE.

Policy	RMSE (m)	Final err (m)	Energy saved (%)
Baseline	1.72 ± 0.70	1.81 ± 0.78	—
All (4 beacons)	0.77 ± 0.27	0.47 ± 0.43	0.00
Adaptive (ours)	0.85 ± 0.34	0.68 ± 0.20	49.9

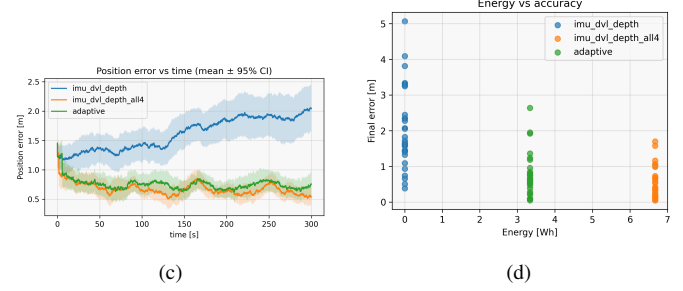


Fig. 2. Adaptive switching cuts acoustic energy by using fewer beacons (c) while keeping final and time-varying position errors close to the always-on 4-beacon baseline(d).

V. CONCLUSION AND FUTURE WORK

This work demonstrates a practical adaptive modem management approach that explicitly trades observability and uncertainty against acoustic energy using SOC-aware scoring, feasibility gating, and hysteresis. In Monte Carlo evaluation, adaptive switching achieved almost 50% energy reduction versus all-four beacons with a modest increase in positioning RMSE. Ongoing and future work will focus on detailed energy modelling and extending scores with channel-quality (SNR/multipath indicators).

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